Meaning in Context: Adult's Understanding of English Bare Plurals

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The "multiple" reading of bare plurals: Semantics or pragmatics?

(Tieu et al., 2020)

"Emily fed pigs."

= Emily fed one or more pigs.

⇝ Emily fed more than one pig.

The Implicature Account

(Sauerland et al. 2005; Spector 2007; Tieu et al., 2020)

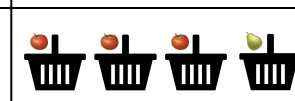
- In English, bare plurals semantically mean "one or more."
- Adults infer the "more than one" reading via a scalar implicature.

If the "multiple" reading arise from implicature, adults should be able to cancel the reading given appropriate contexts.

Relevance of the "multiple" meaning of bare plurals

Relevance	Whether plurals mean "more than one" depends on how *relevant quantity* is in the discourse (Grimm, 2013).
	Relevance (QUD) even boosts children's scalar implicature computation (Skordos & Papafragou, 2016).

Can relevance drive adults to access both readings of bare plurals?

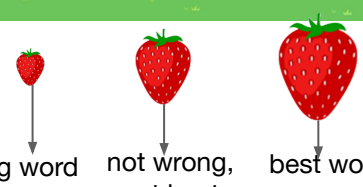
Critical Utterance	Exp. 1 No overt QUD	Exp. 2 - QUD manipulated	
		Quantity Condition	Type Condition
"This basket has pears."			

Predictions:

- Exp1 - Adults should reject multiple-reading use because of implicature (see Tieu et al., 2020).
- Exp2 - Adults should reject multiple-reading use when QUD warrants it (i.e., when quantity, but not object type, is relevant).

Exp. 1: Plurals with single objects - No overt QUD

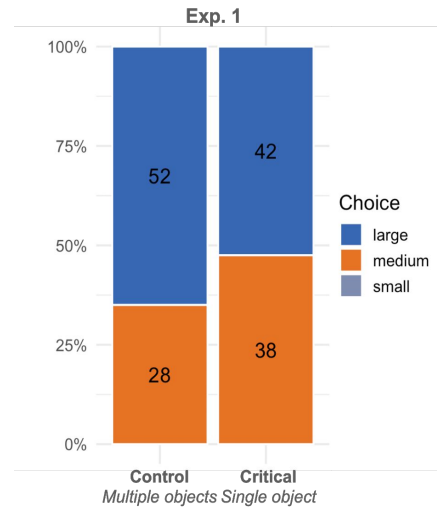
Task: Reward Miss Bunny with a strawberry



wrong word (Katsos & Bishop, 2011) | not wrong, not best | best word

Single-object (Critial) | Three-object (TRUE) | Proper-name (Fillers)

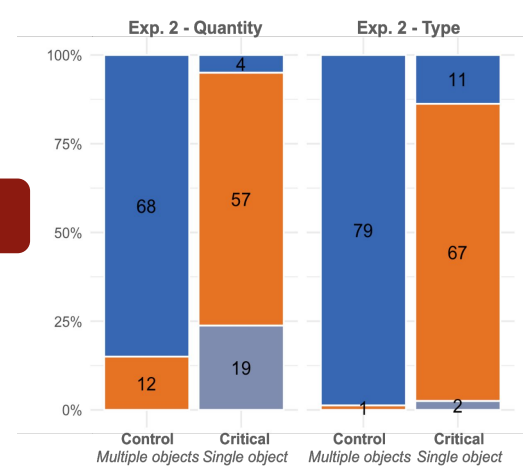
When the context does not raise a quantity-related QUD, adults are equally likely to accept "one or more" and "more than one" interpretations of plurals.



Exp. 2: Plurals with single objects - QUD manipulated

Quantity Condition
"More than one" is *relevant* to identifying the difference

Type Condition
"More than one" is *irrelevant* to identifying the difference



The "multiple" reading of bare plurals are contextual-dependent: Adults tolerated the usage of bare plurals more when the quantity of the referent was less relevant.

Conclusion

- The multiple reading is not the semantics of bare plurals in English.
- Such reading is more readily computed when the context raised a quantity-related QUD.

References

Grimm, S. (2013). Plurality is distinct from number-neutrality. *Proceedings of NELS*, *41*, 247–258. **Katsos, N., & Bishop, D. V. M. (2011).** Pragmatic tolerance: Implications for the acquisition of informativeness and implicature. *Cognition*, *120*(1), 67–81. https://doi.org/10.1016/j.cognition.2011.02.015 **Sauerland, U., Anderssen, J., & Yatsushiro, K. (2005).** The plural is semantically unmarked. *Linguistic Evidence*, *4*, 409–430. **Skordos, D., & Papafragou, A. (2016).** Children's derivation of scalar implicatures: Alternatives and relevance. *Cognition*, *153*, 6–18. https://doi.org/10.1016/j.cognition.2016.04.006 **Spector, B. (2007).** Aspects of the Pragmatics of Plural Morphology: On Higher-Order Implicatures. In U. Sauerland & P. Stateva (Eds.), *Presupposition and Implicature in Compositional Semantics* (pp. 243–281). Palgrave Macmillan UK. https://doi.org/10.1057/9780230210752_9 **Tieu, L., Bill, C., Romoli, J., & Crain, S. (2020).** Testing theories of plural meanings. *Cognition*, *205*, 104307. https://doi.org/10.1016/j.cognition.2020.104307